

Saamil Singhal

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EDUCATION

Vellore Institute of Technology, Bhopal

Bhopal, India

B.Tech in Computer Science Engineering (Artificial Intelligence & Machine Learning)

2023 – 2027

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems, Cloud Computing

EXPERIENCE

Bharat Heavy Electricals Limited (BHEL)

Haridwar, India

Full-Stack Developer Intern

May 2026 – June 2026

- Designed a **relational database schema (SQLite/PostgreSQL)** and developed a scalable **Node.js/Express** backend with **React** frontend for the Smart Material & Inventory Management Platform (SMIMP), building production-ready **REST APIs** and backend workflows.
- Implemented a **predictive forecasting engine** to automate material demand analysis, exposing secure REST API endpoints and optimizing backend data processing workflows.
- Owned end-to-end development including database design, API development, debugging, deployment, testing, Git-based version control, and collaboration with stakeholders to deliver reliable production-ready features.

PROJECTS

Welth - AI-Powered Finance Management Platform

Dec 2025 – Jan 2026

Next.js, React, Prisma, Supabase, Tailwind CSS

- Built and deployed a scalable **AI-powered** personal finance platform with authentication, PostgreSQL, REST APIs, and modular backend workflows to help users manage spending and budgets.
- Designed relational database schemas and implemented secure authentication, backend services, and API-driven workflows using **Prisma ORM** and **Supabase PostgreSQL**.
- Engineered AI-powered financial insights by processing structured financial datasets through scalable backend services and intelligent recommendation workflows.

VoxRAG - AI Video RAG Platform & Evaluation Harness

June 2026 – Aug 2026

Python, LangChain, Mistral AI, ChromaDB, Ragas, Streamlit

[\[GitHub\]](#) · [\[Live Demo\]](#)

- Architected a **RAG pipeline** using **LangChain LCEL** and session-isolated **ChromaDB** collections to deliver context-aware chat with **Mistral AI** and prompt-injection defenses.
- Built a **Ragas**-based evaluation harness tracking Context Precision, Faithfulness, and Answer Relevancy, and hardened the pipeline with exponential backoff against transient API failures.
- Solved HTTP 403 cloud-IP blocks with a caption-first YouTube ingestion strategy and auto-translation fallback, deploying the verified system on **Streamlit Cloud** with **72/72 passing tests**.

Stock Market Trend Predictor

Apr 2025 – Jun 2025

Python, TensorFlow, Pandas, NumPy, Keras

[\[GitHub\]](#) · [\[Live Demo\]](#)

- Built an end-to-end forecasting system using **LSTM** models on live **yFinance API** data to predict market trends.
- Experimented with multiple forecasting architectures, evaluating performance against historical market data.
- Deployed an interactive **Streamlit** dashboard, shipping the model as a usable, live analytics tool.

SKILLS

Languages: Java, Python, SQL

Computer Science Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOP), Computer Networks, Operating Systems, Database Management Systems (DBMS), Complexity Analysis

Cloud & DevOps: Cloud Computing, Cloud Operations, Docker (Basics), Kubernetes (Basics), AWS (Basics), Observability, Metrics

Development: Node.js, Express.js, Flask, FastAPI, REST APIs, ShadCN UI, Authentication, Clerk, Inngest, React, NextJS, HTML, CSS

Databases: PostgreSQL, SQLite, MongoDB, Prisma ORM, SQL

AI & Data Science: Generative AI, RAG, LangChain, LLM Evaluation, Ragas, Prompt Engineering, Mistral AI, ChromaDB, Scikit-learn, Pandas, NumPy

Developer Tools: Git, GitHub, GitHub Actions, VS Code, Streamlit

CERTIFICATIONS & ACHIEVEMENTS

Certifications: Java Foundations Associate (Oracle) | Agentic AI Foundations (Oracle) | Google IT Support Professional Certificate

Leadership & Activities: Core Team Member at GDGC Design & Content Team, VIT Bhopal | Marketing Intern at Ekatva Foundation (Drove Donation Outreach) | 100 Days of Code Challenge (Placement Cell)